

Injective endomorphisms of algebraic and analytic sets

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Abstract . We prove that every injective endomorphism of an affine algebraic variety over an algebraically closed field of characteristic zero is an automorphism . We also construct an analytic curve in $\mathbb{C}^6$ and its holomorphic bijection which is not a biholomorphism .

Introduction . The Zariski Main Theorem (see e . g . [4]) asserts that

every injective morphism of normal algebraic varieties over an algebraically closed field of characteristic zero is an automorphism onto its open image . For arbitrary complex algebraic varieties regularity of the inverse of a regular biholomorphic bijection is ensured by the Serre Theorem [7 , 9] . The simple example $\mathbb{C} \ni t \rightarrow (t^2, t^3) \in \{x^3 - y^2 = 0\}$ shows that in general , bijective morphisms are not isomorphisms . From the example $\{xy - 1 = 0\} \ni (u, v) \rightarrow (u, 0) \in \{y = 0\}$ we see that an injective morphism of non - singular algebraic sets may not be surjective .

Grothendieck proved in [5 , Prop . 17 . 9 . 6] a counterpart of the Zariski

Main Theorem for the category of S - preschemes , where S is a fixed pre - scheme . His theorem says that every injective S - endomorphism of an S - prescheme of finite presentation is an automorphism . Note that his theorem is unapplicable to endomorphisms of algebraic sets .

Our main result gives a geometric counterpart of the Grothendieck result ; Theorem 2 . 2 says that every injective endomorphism of an affine algebraic variety over an algebraically closed field $\mathbf{k}$ of characteristic zero is an automorphism . This generalizes the well - known result of Bia łyński - Birula and Rosenlicht [2] saying that every injective polynomial transformation of $\mathbf{k}^n$ is a polynomial automorphism .

line - N_{ote} that surjectivity of an injective endomorphism of an affine algebraic variety was earlier proved by Ax [1] . In the proof Ax applied the so - called

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“transfer principle” and the metamathematical notion of an “elementary formula”. A topological proof of Ax’s Theorem was given by Borel in his unpublished paper [3].

One can also consider Theorem 2.2 as an analogue of the Clements – Osgood Theorem on analyticity of the inverse of a holomorphic bijection of a complex manifold. In Section 3 we show that there is no counterpart of Theorem 2.2 for the category of analytic sets. In order to obtain a suitable counterexample we construct an irreducible one-dimensional complex analytic space and its holomorphic but not biholomorphic bijection and next we prove that this analytic space is biholomorphic with an analytic curve in $\mathbb{C}^6$.

1. Dominating regular mappings with finite fibres. Let $F : V \rightarrow W$ be a regular dominating mapping of irreducible affine algebraic varieties over $\mathbf{k}$, i.e. $F(V) = W$. Then the induced homomorphism $F_* : \mathbf{k}[W] \rightarrow \mathbf{k}[V]$ of rings of regular functions is injective, so it has a unique extension to a monomorphism of fields of rational functions, also denoted by $F_* : \mathbf{k}(W) \rightarrow \mathbf{k}(V)$. If we assume additionally that $\dim V = \dim W$ then the field $\mathbf{k}(V)$ is a finite extension of the field $F_*(\mathbf{k}(W))$; we put $d(F) = [\mathbf{k}(V) : F_*(\mathbf{k}(W))]$. It is well known that $d(F)$ is equal to the number of points in the generic fibre of F (see e.g. [4]).

In some cases we have more precise information about the number $d(F)$:

THEOREM 1.1. *Let V and W be normal affine varieties and $\dim V = \dim W$. If $F : V \rightarrow W$ is a regular dominating mapping with finite fibres*

$$\text{then } d(F) = \max\{\#F^{-1}(y) : y \in W\}.$$

For the proof we need the following

LEMMA 1.2. *Let V and W be irreducible affine algebraic varieties such that $\dim V = \dim W$ and V is normal. Assume that $F : V \rightarrow W$ is a regular dominating mapping with finite fibres. Then, for every $y \in W$ with $F^{-1}(y) = \{x_1, \dots, x_m\} \subset V$, there exists a function $f \in \mathbf{k}[V]$ which is integral over $F_*(\mathbf{k}[W])$ and $f(x_i) \neq f(x_j)$ for $i \neq j$.*

PROOF. By [4, Corollary 1, p. 136] we can find a normal affine algebraic variety W' , a regular injective mapping $F' : V \rightarrow W'$ and a regular finite mapping $G : W' \rightarrow W$ such that $F = G \circ F'$.

Then the points $\{F'(x_j) : j = 1, \dots, m\} \subset W'$ are pairwise different, so there exists $g \in \mathbf{k}[W']$ such that $g(F'(x_i)) \neq g(F'(x_j))$ for $i \neq j$. Since G is finite, the ring $\mathbf{k}[W']$ is integral over the ring $G_*(\mathbf{k}[W])$. Therefore $f = g \circ F' = F'_*(g)$ has all desired properties.

PROOF OF THEOREM 1.1. Take a $y \in W$ and the function $f \in \mathbf{k}[V]$ from Lemma 1.2 which is integral over the ring $F_*(\mathbf{k}[W])$ and separates the

points in the fibre $F^{-1}(y) = \{x_1, \dots, x_m\} \subset V$.

In the extension of integral rings $F_*(\mathbf{k}[W]) \subset \mathbf{k}[V]$ the ring $F_*(\mathbf{k}[W])$ is normal. One can easily verify that $f \in \mathbf{k}[V]$ is algebraic over the field $F_*(\mathbf{k}(W))$ and the coefficients of its minimal equation of degree d over $F_*(\mathbf{k}(W))$ lie in $F_*(\mathbf{k}[W])$. Since all functions from $F_*(\mathbf{k}[W])$ are constant on the set $F^{-1}(y)$ whose elements are separated by f , we have $m \leq d \leq d(F)$.

$$\text{Henced}(F) = \max\{\#F^{-1}(y) : y \in W\}.$$

2. Bijectivity implies biregularity. Assume that all varieties occurring in this section are defined over a field $\mathbf{k}$ (algebraically closed of characteristic zero).

LEMMA 2.1. *Let V be an affine algebraic variety and let $\widehat{V} = V_1 \cup \dots \cup V_s$ be its decomposition into irreducible components. Denote by $\pi_i : \widehat{V}_i \rightarrow V_i$ a normalization of V_i for $i = 1, \dots, s$. Then*

$$M_V = \{f : V \rightarrow \mathbf{k} : f \circ \pi_i \in \mathbf{k}[\widehat{V}_i] \text{ for } i = 1, \dots, s\}$$

is a noetherian $\mathbf{k}[V]$ -module (under the standard operations).

PROOF. Since $\mathbf{k}[\widehat{V}_i]$ is a noetherian $\mathbf{k}[V_i]$ -module, it is also a noetherian $\mathbf{k}[V]$ -module for every $i = 1, \dots, s$ (under the multiplication $rf = (r \circ \pi_i)f$ for $r \in \mathbf{k}[V]$, $f \in \mathbf{k}[\widehat{V}_i]$.) The mapping

$$M_V \ni f \rightarrow (f \circ \pi_1, \dots, f \circ \pi_s) \in \bigoplus_{i=1}^s \mathbf{k}[\widehat{V}_i]$$

is a monomorphism of $\mathbf{k}[V]$ -modules. Therefore M_V is a noetherian $\mathbf{k}[V]$ -module.

Let V be an irreducible affine algebraic variety over $\mathbf{k}$ and let $\widehat{V}$ together with a finite surjective regular and birational mapping $\pi : \widehat{V} \rightarrow V$ be a normalization of V .

By the universal property of normalization, for a given regular endomorphism $F : V \rightarrow V$ there exists a unique endomorphism $\widehat{F} : \widehat{V} \rightarrow \widehat{V}$ such that $\pi \circ \widehat{F} = F \circ \pi$. We have

LEMMA 2.2. (i) *If F is injective then $\widehat{F}$ is injective.*

(ii) *If $\widehat{F}$ is surjective then F is surjective.*

PROOF. (i) Assume that F is injective. First observe that $\widehat{F}$ is dominating. Since F is dominating and $\pi : \widehat{V} \rightarrow V$ is surjective, we see that $F \circ \pi : \widehat{V} \rightarrow V$ is dominating. Since π is closed, $\pi(\widehat{F}(\widehat{V})) = \pi(\widehat{V}) = F(\pi(\widehat{V})) = V$. If $\widehat{F}(\widehat{V}) \subsetneq \widehat{V}$ then by irreducibility of $\widehat{V}$ we would have $\dim V = \dim \pi(\widehat{F}(\widehat{V})) \leq \dim \widehat{F}(\widehat{V}) < \dim \widehat{V}$, a contradiction.

Therefore $\widehat{F}(\widehat{V}) = \widehat{V}$. Moreover, $1 = d(F)d(\pi) = d(F \circ \pi) = d(\pi \circ \widehat{F}) =$

$d(\widehat{F})$. Since all fibres of π are finite, the same holds for $\widehat{F}$. By Theorem 1.1, $\{y \in \widehat{V} : \#\widehat{F}^{-1}(y) \neq 1\} = \{y \in \widehat{V} : \widehat{F}^{-1}(y) = \emptyset\}$, which means that $\widehat{F}$ is injective.

(ii) If $\widehat{F}$ is surjective then $\pi \circ \widehat{F} = F \circ \pi$ is also surjective, and hence so is F .

Our main result is

THEOREM 2.2. *Let V be an affine algebraic variety and let $F : V \rightarrow V$ be a regular mapping. Then the following statements are equivalent :*

(i) F is injective.

(ii) F is a bijection of V .

(iii) F is an automorphism of V .

P r o o f . The implication (i) $\Rightarrow$ (ii) is true by the Ax Theorem and (iii) implies (i).

(ii) $\Rightarrow$ (iii). Assume that V is an algebraic subset of $\mathbf{k}^n$ for some $n \in \mathbb{N}$ and $V = V_1 \cup \dots \cup V_s$ is the decomposition of V into irreducible components. Denote by $\pi_i : \widehat{V}_i \rightarrow V_i$ a normalization of V_i for $i = 1, \dots, s$. Then, by Lemma 2.1,

$$M_V^n = \bigoplus_{i=1}^s M_{V_i}^n = \{G : V \rightarrow \mathbf{k}^n :$$

$$G \circ \pi_i : \widehat{V}_i \rightarrow \mathbf{k}^n \text{ is regular for } i = 1, \dots, s\}$$

is a noetherian $\mathbf{k}[V]$ -module.

Without loss of generality we can assume that $F(V_i) \subset V_i$ for $i = 1, \dots, s$. Let F_i denote the restriction of F to V_i . By Lemma 2.2, the Ax Theorem and Zariski's Main Theorem, every mapping $\widehat{F}_i$ is an automorphism of $\widehat{V}_i$. Therefore, for every $\nu \in \mathbb{N}$ and every $i \in \{1, \dots, s\}$, the mapping $F^{-\nu} \circ \pi_i = \pi_i \circ (\widehat{F}_i)^{-\nu}$ is regular, so $F^{-\nu} \in M_V^n$.

For every $m \in \mathbb{N}$ we put $S_m = \mathbf{k}[V]F^{-1} + \dots + \mathbf{k}[V]F^{-m}$. Then the S_m are submodules of the noetherian $\mathbf{k}[V]$ -module M_V^n and $S_1 \subset S_2 \subset S_3 \subset \dots$. Therefore there exists an $l \in \mathbb{N}$ such that $S_l = S_{l+1}$ and we can write

$$F^{-(l+1)} = \sum_{j=1}^l r_j F^{-j} \quad \text{for some } r_1, \dots, r_l \in \mathbf{k}[V].$$

$$j = 1$$

Thus the mapping

$$F^{-1} = F^{-(l+1)} \circ F^l = \sum_{j=1}^l (r_j \circ F^l) F^{l-j}$$

is regular and the proof is complete.

R e m a r k . Note that in [8] Matsumura and Monsky gave examples of hypersurfaces in $\mathbb{C}^n$ having no automorphisms (hence no regular injective endomorphisms) apart from the identity .

3 . Counterexample in the analytic case . We first construct an analytic space and a holomorphic bijection of this space onto itself which is not a biholomorphism .

EXAMPLE 3 . 1 . Let X denote the topological space obtained from $Y = \mathbb{C} \setminus (\mathbb{Z} \setminus \mathbb{N})$ by gluing every pair of points l and $l + 1/2$ into one point $a_l \in X$ for $l \in \mathbb{N}$. Let $\pi : Y \rightarrow X$ denote the canonical mapping defining this quotient Hausdorff space .

We have $X = X_0 \cup \{a_0, a_1, \dots\}$, where $X_0 = \mathbb{C} \setminus (\mathbb{Z} \cup \{k + 1/2 : k \in \mathbb{N}\})$ is an open subset of X . Moreover ,

$$U_l = \{a_l\} \cup (B(l, 1/5) \setminus \{l\}) \cup (B(l + 1/2, 1/5) \setminus \{l + 1/2\})$$

is a neighbourhood of a_l in X . ($B(z, r)$ denotes the open ball with centre

$$z \in \mathbb{C} \text{ and radius } r > 0.)$$

We define a complex structure on X by the family of homeomorphisms :

$$\begin{aligned} \text{id}_{X_0} : X_0 &\rightarrow X_0, \\ \phi_l : U_l &\rightarrow (B(0, 1/5) \times \{0\}) \cup (\{0\} \times B(0, 1/5)), \quad l \in \mathbb{N}, \end{aligned}$$

where

$$\phi_l(z) = \begin{cases} 0 & \text{if } z = a_l, \\ (z - l, 0) & \text{if } z \in B(l, 1/5) \setminus \{l\}, \\ (0, z - l - 1/2) & \text{if } z \in B(l + 1/2, 1/5) \setminus \{l + 1/2\}. \end{cases}$$

They define on X the structure of a one - dimensional irreducible complex space with singularities exactly at the points a_l ; we write $\text{Sing } X = \{a_l : l \in \mathbb{N}\}$.

$$l \in \mathbb{N}\}.$$

Define the mapping $\Phi : X \rightarrow X$ by the formula

$$\Phi(x) = \begin{cases} x + 1 & \text{if } x \in X_0 \setminus \{-1/2\}, \\ a_{l+1} & \text{if } x = a_l, \quad l \in \mathbb{N}, \\ a_0, & \text{if } x = -1/2. \end{cases}$$

Then Φ is a holomorphic bijection of X onto X but not an open mapping (it is not open at the point $-1/2$). Therefore $\Phi^{-1} : X \rightarrow X$ is not holomorphic .

The complex space X constructed above can be “ well ” embedded in $\mathbb{C}^6$:

PROPOSITION 3 . 2 . *There exists a biholomorphic mapping of X onto an irreducible analytic curve $V \subset \mathbb{C}^6$.*

P r o o f . Since X is one - dimensional , it is a Stein space (see [6 , Theorem IX . B . 1 0]) . Therefore there exists a holomorphic homeomorphism θ of X onto an irreducible analytic curve in $\mathbb{C}^4$ (see [6 , Theorem VII . C . 1 0]) .

Observe that the set $\{x \in X : \text{rank}_x \theta < \text{dim}_x X\}$ is contained in $\pi(\{x \in X_0 : d_x(\theta \circ \pi) = 0\} \cup \text{Sing } X)$, so it is at most countable . (Following [6], $\text{dim}_x X$ denotes the tangential dimension of X at x ; $\text{rank}_x \theta$ denotes the rank of the induced linear mapping of Zariski tangent spaces .)

Since $\text{dim}_x X \leq 2$ for every $x \in X$, applying at most twice Lemma VII . C . 1 1 of [6] we can find a holomorphic homeomorphism Ψ of X onto an irreducible analytic curve V in $\mathbb{C}^6$ such that $\text{rank}_x \Psi = \text{dim}_x X$ for every

$$x \in X.$$

We claim that Ψ is a biholomorphism of X onto V . Indeed , take an $x_0 \in X$ and a complex manifold M of dimension $\text{dim}_{x_0} X$ such that $M \cap X$ is a neighbourhood of x_0 in X . Let $\tilde{\Psi}$ be a holomorphic extension of Ψ to M . Then $\text{rank}_{x_0} \tilde{\Psi} = \text{dim}_{x_0} X$. Since the function $M \ni x \rightarrow \text{rank}_x \tilde{\Psi} \in \mathbb{N}$

is lower semicontinuous, shrinking M if necessary we conclude that $\tilde{\Psi} : M \rightarrow \mathbb{C}^6$ is a biholomorphism onto its image. Then $\tilde{\Psi}^{-1}_{(N \cap \Psi(X))} = M \cap X$, i.e. $\tilde{\Psi}^{-1} = \Psi^{-1}$ in aneighbourhood of $\Psi(x_0)$ in $V = \Psi(X)$. Therefore Ψ^{-1} is holomorphic at $\Psi(x_0) \in V$, and this proves that Ψ is a biholomorphic mapping of X onto V , as desired .

The above constructions give

EXAMPLE 3 . 3 . The mapping $F = \Psi \circ \theta \circ \Psi^{-1} : V \rightarrow V$ is a holomorphic homeomorphic bijection of the irreducible analytic curve $V \subset \mathbb{C}^6$ onto itself with F^{-1} not holomorphic .

In this context the following question seems to be interesting :

PROBLEM . Is each holomorphic bijective self - transformation of an algebraic set necessarily biholomorphic ?

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